

SCI-NOI v0.1

SCI-NOI v0.1 r0.5 Engineering Conformance Specification

Implementation-blind Stage B draft

Scientific owner: Grant Wilson

SCI-NOI_ENGINEERING_CONFORMANCE v0.1/draft-r0.5

Conditional owner-freeze candidate; not frozen. No implementation conformity, generator validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Scientific owner: Grant Wilson

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Status: implementation-blind proposed scientific-owner freeze candidate; scientifically freezeable conditional; not frozen.

Normative authority: the six ordered modules in SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.5, binding-file SHA-256 aa59ecaaaa149e2990d07623563d90af76c7b3084ee37c497a06e17ebf0fe213. This ECS supplies prospective evidence procedures only and authors no science.

1. Evidence identity and result vocabulary

Every result shall identify exact product, method, parent, observation, TolTEC array, plan, generation, lifecycle, and named use. Review compares serialized scientific identities and typed states, never filenames, shapes, generic flags, approximate WCS, display decimals, or undocumented defaults.

Method identity, parent/boundary source closure, changed use-profile authority, plan resolution, numerical realization, response-bearing realization, random- generator validation, empirical calibration, implementation conformity, validation, and production authorization shall be reported separately as pass, fail, unavailable, or not applicable with exact cause.

2. Request lifecycle and cardinality evidence

A not_requested fixture shall have request axis not_requested, no GEN method or eligibility proposition, no request/effective-plan identity, no resolved assignment-design identity, N_requested=0, and no downstream object.

An explicitly_disabled fixture shall retain exact request and effective-plan identities, effective state disabled, and disabling owner, policy, and cause, while having no resolved assignment-design identity, N_requested=0, member, UNC, or STD. Evidence shall reject aliasing the two zero-cardinality states.

An enabled-success fixture shall have positive integer request and N_resolved=N_requested>0. If any requested assignment exhausts its cap, the complete design shall fail and publish no smaller successful ensemble. Rejected candidates shall appear only as construction events.

3. Frozen-parent and lossless-rational evidence

Evidence shall expose retained PTC occurrence, detector/network, coefficient family/value/QC, exact frozen MAP product/generation, G_pi, gamma_i family/generation/payload, exact parent contribution, lossless canonical rational, representation source/digest, projection, unsigned denominator, numerical coverage_cut, WCS/support, response, and application count.

Decoding the rational shall equal the exact parent a_pi bit-for-value under the declared lossless representation. Evidence shall reject display-text rounding, independent quantization, re-estimation, undocumented decimal precision, and another coefficient generation. The rational may determine NOI balance/design identity only; it shall not replace MAP numerical arithmetic.

Mandatory mismatch fixture:

- Fact | Exact value
- Frozen-parent contribution and lossless rational | 100000000000000001/100000000000000000
- Rounded display text | 0.1
- Total and tolerance | $T_h=1/1$, $\tau_h=1/10$

- Required exact decision | reject, because the lossless value is above the boundary
- Prohibited display-derived decision | admit by treating rounded 0.1 as equality

Missing lossless identity makes design resolution unavailable.

4. Exact design and arithmetic evidence

Every design record shall include exact per-stratum admitted sets A_h , canonical stratum order and concatenation, $A = \text{product}_h A_h$, stable detector order, alphabet, ideal probabilities, duplicates/complements, all counts, ranks/null spaces, reconstruction identity, lifecycle, completion/failure, and preregistration before assignment or output inspection.

Canonical ASCII rationals shall reject nonpositive denominators, unreduced pairs, noncanonical zero, nonpositive active mass, and thresholds outside $[0,1)$. Exact accumulation and integer cross multiplication shall reproduce every mass, numerator, total, and admitted-set decision without floating epsilon, tolerance relaxation, or best-failed selection. Equality, one exact rational step above, one step below, and reordered wide-magnitude accumulation fixtures shall produce their prescribed exact decisions.

5. Target probability law and deterministic replay evidence

Evidence shall represent separately:

1. the scientific Uniform(A) randomization measure;
2. the declared ideal independent symmetric candidate-bit/key-selection law;
3. the deterministic replay map $F(K, \text{plan})$.

Exact enumeration or analytic proof shall establish first-accepted conditional uniformity over each A_h and the canonical product A . The plan shall bind generator identity/version, opaque key bytes, namespace, serialization, observation, array, stratum, member, attempt counter, detector order, and domain separation. Replays varying scheduling, process/worker count, traversal, container layout, and persistence shall reproduce the same assignment sequence.

Review shall record that replay equality establishes realized identity only. A single realized sequence and replay success shall not pass iid-independence, generator-quality, random-generator-validation, or estimator-uncertainty gates. Those gates remain unavailable pending separately named methods and evidence.

6. Design-ignorability and member-admission evidence

Before assignment, evidence shall serialize every binding predicate in R_b or identify its exact invariance proof. For every enumerated e in A , a deterministic predicate shall return the same outcome, or a probabilistic proof shall establish EQ-017. Output amplitude, sign, source imprint, morphology, tail value, and assignment-dependent QC shall be advisory and shall not alter base membership.

Mandatory fixtures shall show:

- a parent/plan-fixed predicate preserving the target law;
- an advisory output diagnostic changing value without changing membership;
- a sign- or amplitude-dependent exclusion violating EQ-017 and making the base method unavailable; and
- a failed design not being regenerated under the same method identity until a favorable all-pass ensemble appears.

A scientifically required assignment-dependent gate shall fail the base-method identity check and require a new selection-conditioned target law, estimator, weights, uncertainty, and claim. No evidence may report the unconditioned Uniform(A) law after output-dependent selection.

7. Conditional UNC realization and information evidence

For a realized initial UNC, evidence shall prove $A_{\text{UNC}}=B_{\text{resolved}}$ and $N_{\text{admitted}}=N_{\text{completed}}=N_{\text{resolved}}=N_{\text{requested}}>0$, every positive exact member decision, EQ-017, and one common domain. Estimator reproduction shall then use the exact known-zero-center formula with divisor N_{resolved} , no finite- ensemble centering, and no B-1.

For failed or unavailable UNC, evidence shall retain actual counts, member and ensemble causes, and any inequalities. It shall prove absence of estimator, survivor subset, renormalization, and changed divisor. A completed-but-policy- unavailable member and an assignment-dependent failure are mandatory unavailable-state fixtures.

Counts, duplicate/orbit facts, r_{sign} , r_{map} , estimator uncertainty, and effective information shall remain separate. An effective-information product shall state estimator, target, domain, dependence model, and calculation or be unavailable. Deterministic replay supplies none of these quantities by itself.

8. STD, profiles, and freeze evidence

STD evidence shall bind independently realized `m_MAP` and compatible `Vhat_cond`, reproduce the scale and unit-one product, and emit only the bounded claim. Fixed-scale and full-response fixtures shall remain separate; without separately authorized delta `Vhat_cond`, full response is unavailable. No reciprocal, interpolation, JINC substitution, or significance claim is allowed.

Every immutable `r0.18` profile and Registry/source digest shall reproduce. Each changed use shall bind the exact revised `@2` bytes, owner approval, and new SCI-VAL source/Registry record before evaluation. An old profile or proposed name shall not authorize a changed action; missing authority yields unavailable, not numerical ineligibility or empty success.

`REQUIREMENT_PREDICTION_OWNER_TRACEABILITY.csv` shall cover every stable ID. The superseding freeze manifest shall bind all owner directives, modules, profiles, bindings, views, parity reports, PDFs, and freeze record. This draft may support a conditional scientific-owner freeze disposition, but supplies no current freeze, implementation conformity, random-generator validation, calibration, empirical validation, readiness, or production result.

Appendix A. Exact Requirement/Prediction/Owner Traceability

SCI-NOI-REQ-001

Predictions: SCI-NOI-PRED-011 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: SCOPE|TAX Parity: exact ECS section/review: 1 / operation-role review

SCI-NOI-REQ-002

Predictions: SCI-NOI-PRED-004 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-identity-numerically-unavailable Dependency: PTCB|MAPB Parity: exact ECS section/review: 2 / operator identity review

SCI-NOI-REQ-003

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-101|ODQ-104 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: GRAPH|FILTER Parity: exact ECS section/review: 8 / fixed-relearned review

SCI-NOI-REQ-004

Predictions: SCI-NOI-PRED-002 Owner decisions: ODQ-102B Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: PTCB|DESIGN Parity: exact ECS section/review: 3 / coherence audit

SCI-NOI-REQ-005

Predictions: SCI-NOI-PRED-001 Owner decisions: ODQ-102C Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: DESIGN|MAPB Parity: exact ECS section/review: 3 / mass-balance audit

SCI-NOI-REQ-006

Predictions: SCI-NOI-PRED-015 Owner decisions: SCI-NOI-OWNER-Q-SCOPE-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: SCOPE Parity: exact ECS section/review: 1 / observation-array scope audit

SCI-NOI-REQ-007

Predictions: SCI-NOI-PRED-004 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-identity-numerically-unavailable Dependency: PTCB|MAPB Parity: exact ECS section/review: 2 / sign-placement audit

SCI-NOI-REQ-008

Predictions: SCI-NOI-PRED-004 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: MAPB Parity: exact ECS section/review: 2 / representation and parent audit

SCI-NOI-REQ-009

Predictions: SCI-NOI-PRED-012 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: MAPB|GRAPH Parity: exact ECS section/review: 2 / response parity review

SCI-NOI-REQ-010

Predictions: SCI-NOI-PRED-015|SCI-NOI-PRED-016 Owner decisions: ODQ-102D|SCI-NOI-OWNER-Q-102D-01 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / population-state audit

SCI-NOI-REQ-011

Predictions: SCI-NOI-PRED-015|SCI-NOI-PRED-016 Owner decisions: SCI-NOI-OWNER-Q-102D-01|r0.5-final-directive Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5|f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / edge-state audit

SCI-NOI-REQ-012

Predictions: SCI-NOI-PRED-014 Owner decisions: SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / rejection mechanics audit

SCI-NOI-REQ-013

Predictions: SCI-NOI-PRED-013 Owner decisions: r0.3-architecture-basis|SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN|LIFE Parity: exact ECS section/review: 3 / universal design identity review

SCI-NOI-REQ-014

Predictions: SCI-NOI-PRED-014|SCI-NOI-PRED-021 Owner decisions: SCI-NOI-OWNER-Q-102D-01|r0.5-final-directive Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5|f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 5 / joint-law and key-identity review

SCI-NOI-REQ-015

Predictions: SCI-NOI-PRED-005 Owner decisions: SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact-repaired ECS section/review: 4 / duplicate-complement audit

SCI-NOI-REQ-016

Predictions: SCI-NOI-PRED-005 Owner decisions: SCI-NOI-OWNER-Q-102D-01|ODQ-105B Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5|272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: DESIGN|UNCT Parity: exact ECS section/review: 5 / count-rank-information review

SCI-NOI-REQ-017

Predictions: SCI-NOI-PRED-003|SCI-NOI-PRED-007 Owner decisions: ODQ-103 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 6 / source-imprint claim review

SCI-NOI-REQ-018

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-023 Owner decisions: ODQ-105A|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved Dependency: LIFE|VALP|SELECT Parity: exact ECS section/review: 6 / assignment-dependent failure and favorable-regeneration review

SCI-NOI-REQ-019

Predictions: SCI-NOI-PRED-013 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: LIFE Parity: exact ECS section/review: 5 / retry-replacement identity audit

SCI-NOI-REQ-020

Predictions: SCI-NOI-PRED-009 Owner decisions: ODQ-109 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: LIFE Parity: exact ECS section/review: 5 / persistence plan review

SCI-NOI-REQ-021

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-110A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-boundary-numerically-unavailable Dependency: FILTER Parity: exact ECS section/review: 8 / external transform parity review

SCI-NOI-REQ-022

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-110B|ODQ-110C Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-boundary-numerically-unavailable Dependency: FILTER|LIFE Parity: exact ECS section/review: 8 / generation separation review

SCI-NOI-REQ-023

Predictions: SCI-NOI-PRED-006|SCI-NOI-PRED-007 Owner decisions: ODQ-105B|ODQ-106 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-estimand Dependency: UNCT Parity: exact ECS section/review: 6 / UNC completeness review

SCI-NOI-REQ-024

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-019|SCI-NOI-PRED-023 Owner decisions: ODQ-105A|ODQ-111|r0.4-final-directive|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-method-r0.5-profile-evaluation-unavailable Dependency: UNCT|VALP|SELECT Parity: exact ECS section/review: 7 / conditional-realization and design-ignorability audit

SCI-NOI-REQ-025

Predictions: SCI-NOI-PRED-002|SCI-NOI-PRED-006|SCI-NOI-PRED-019 Owner decisions: ODQ-105B|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423 Approval: approved-estimand Dependency: UNCT Parity: exact ECS section/review: 6 / estimator normalization reproduction

SCI-NOI-REQ-026

Predictions: SCI-NOI-PRED-007|SCI-NOI-PRED-009 Owner decisions: ODQ-105B|r0.3-architecture-basis
Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: UNCT Parity: exact ECS
section/review: 6 / product-name-unit-claim review

SCI-NOI-REQ-027

Predictions: SCI-NOI-PRED-009 Owner decisions: ODQ-105B Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: UNCT
Parity: exact ECS section/review: 6 / uncertainty-information separation

SCI-NOI-REQ-028

Predictions: SCI-NOI-PRED-005 Owner decisions: ODQ-106 Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency:
UNCT|TAX Parity: exact ECS section/review: 6 / representation-role review

SCI-NOI-REQ-029

Predictions: SCI-NOI-PRED-006 Owner decisions: ODQ-107|r0.3-no-new-reciprocal-decision Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: legacy-approved-r0.3-successor-unavailable Dependency:
UNCT|VALP Parity: typed-unavailability ECS section/review: 7 / reciprocal gate review

SCI-NOI-REQ-030

Predictions: SCI-NOI-PRED-017 Owner decisions: ODQ-108|ODQ-102A Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval:
approved-method-numerically-unavailable Dependency: STDT|MAPB|JINCB Parity: exact ECS
section/review: 7 / numerator-scale-parent audit

SCI-NOI-REQ-031

Predictions: SCI-NOI-PRED-017 Owner decisions: ODQ-108 Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: STDT
Parity: exact ECS section/review: 7 / unit-claim review

SCI-NOI-REQ-032

Predictions: SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval:
approved-r0.18-bytes-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS
section/review: 8 / profile projection review

SCI-NOI-REQ-033

Predictions: SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest:
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval:
approved-r0.18-bytes-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS
section/review: 8 / GEN-input profile review

SCI-NOI-REQ-034

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-011|SCI-NOI-PRED-023 Owner decisions:
ODQ-111|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval:
approved-r0.18-bytes-r0.5-successor-unavailable Dependency: VALP|REG_BIND|SELECT Parity:
exact-boundary ECS section/review: 6 / assignment-ignorable UNC-member profile review

SCI-NOI-REQ-035

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-011|SCI-NOI-PRED-023 Owner decisions: ODQ-111|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-r0.18-bytes-r0.5-successor-unavailable Dependency: VALP|REG_BIND|SELECT Parity: exact-boundary ECS section/review: 6 / design-ignorable UNC-ensemble profile review

SCI-NOI-REQ-036

Predictions: SCI-NOI-PRED-011|SCI-NOI-PRED-017|SCI-NOI-PRED-022 Owner decisions: ODQ-111|ODQ-108|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423 Approval: r0.18-approved-r0.5-successor-proposed-unavailable Dependency: VALP|STDT|REG_BIND Parity: exact-boundary ECS section/review: 8 / STD successor and response gate review

SCI-NOI-REQ-037

Predictions: SCI-NOI-PRED-012 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: CONV Parity: exact ECS section/review: 1 / typed-state no-default review

SCI-NOI-REQ-038

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-110A|ODQ-110B|ODQ-110C Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: FILTER|LIFE Parity: exact ECS section/review: 8 / immutable ancestry review

SCI-NOI-REQ-039

Predictions: SCI-NOI-PRED-009 Owner decisions: ODQ-102C|ODQ-105B|ODQ-107 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: TAX|UNCT|STDT Parity: exact ECS section/review: 7 / role-promotion prohibition

SCI-NOI-REQ-040

Predictions: SCI-NOI-PRED-011 Owner decisions: ODQ-111|r0.4-final-directive|r0.5-final-directive Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: r0.18-approved-and-bound-r0.5-successors-proposed-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS section/review: 8 / profile immutability and revised-successor review

SCI-NOI-REQ-041

Predictions: SCI-NOI-PRED-001 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: COVER Parity: exact ECS section/review: 8 / module binding review

SCI-NOI-REQ-042

Predictions: SCI-NOI-PRED-017|SCI-NOI-PRED-024 Owner decisions: r0.3-architecture-basis|r0.5-final-directive Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved Dependency: SCOPE|COVER Parity: exact ECS section/review: 8 / claim-ceiling and generator-validation review

SCI-NOI-REQ-043

Predictions: SCI-NOI-PRED-018|SCI-NOI-PRED-026 Owner decisions: r0.4-final-directive|r0.5-final-directive Owner digest: f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved Dependency: CARD|LIFE Parity: exact ECS section/review: 2 / zero-cardinality lifecycle review

SCI-NOI-REQ-044

Predictions: SCI-NOI-PRED-020|SCI-NOI-PRED-025 Owner decisions: r0.4-final-directive|r0.5-final-directive|r0.4-author-rational-selection Owner digest: f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-author-selection Dependency: DESIGN|CONV|MAPB Parity: exact ECS section/review: 3 / lossless-rational and boundary-fixtue review

SCI-NOI-REQ-045

Predictions: SCI-NOI-PRED-014|SCI-NOI-PRED-021|SCI-NOI-PRED-024 Owner decisions: r0.4-final-directive|r0.5-final-directive Owner digest: f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423|4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-ideal-law-generator-validation-unassessed Dependency: DESIGN|RNGV Parity: exact ECS section/review: 5 / ideal-law versus deterministic-replay review

SCI-NOI-REQ-046

Predictions: SCI-NOI-PRED-009 Owner decisions: r0.4-final-directive Owner digest: f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423 Approval: approved Dependency: UNCT Parity: exact ECS section/review: 7 / effective-information typing review

SCI-NOI-REQ-047

Predictions: SCI-NOI-PRED-022 Owner decisions: r0.4-final-directive Owner digest: f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423 Approval: approved-method-full-response-unavailable Dependency: STDT|RESP Parity: exact ECS section/review: 8 / fixed-scale versus full-response review

SCI-NOI-REQ-048

Predictions: SCI-NOI-PRED-023 Owner decisions: r0.5-final-directive Owner digest: 4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-base-method-selection-invariant Dependency: SELECT|UNCT|VALP Parity: exact ECS section/review: 6 / EQ-017 invariance and output-selection review

SCI-NOI-REQ-049

Predictions: SCI-NOI-PRED-024 Owner decisions: r0.5-final-directive Owner digest: 4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-disposition-B-generator-validation-unassessed Dependency: DESIGN|RNGV Parity: exact ECS section/review: 5 / target-measure ideal-law replay separation review

SCI-NOI-REQ-050

Predictions: SCI-NOI-PRED-025 Owner decisions: r0.5-final-directive Owner digest: 4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved-method-numerically-unavailable Dependency: MAPB|DESIGN|CONV Parity: exact ECS section/review: 3 / lossless parent-lift mismatch-fixtue review

SCI-NOI-REQ-051

Predictions: SCI-NOI-PRED-026 Owner decisions: r0.5-final-directive Owner digest:
4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d Approval: approved Dependency:
LIFE|CARD Parity: exact ECS section/review: 2 / not-requested versus explicitly-disabled review